

Statistical Learning

Linear Models for Regression

- Linear Regression Review
- Training vs. Testing Errors
- Model Selection Criteria and Algorithms

Linear Models for Regression

Regression Models

- Observe a collection of n i.i.d. **training samples**

$$\mathcal{D}_n = \{x_i, y_i\}_{i=1}^n$$

where x_i is a p dimensional vector (predictors, covariates, features, inputs)

$$x_i = (x_{i1}, \dots, x_{ip})^\top$$

and $y_i \in \mathbb{R}$ is a **continuous response** (outcome, output).

- We assume the underlying model $Y = f(X) + \epsilon$
- Estimate f using $\hat{f}$

- $\mathbf{x}_j$ is a n dimensional vector of the j th feature, i.e.,

$$\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{nj})^T$$

- The design matrix $\mathbf{X}$ is $n \times p$:

$$\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p)$$

- $\mathbf{x}_j$ is one column in $\mathbf{X}$

Loss and Risk

- What is a good model fitting?
- Loss function, risk, and empirical risk
- A **loss** function L measures the discrepancy between Y and any function $f(X)$
- **Risk** is the expected loss over the entire population

$$R(f) = E [L(Y, f(X))]$$

- The true function $f(x)$ would minimize this risk.

- In regression, the **squared error loss** is commonly used:

$$L(Y, f(X)) = (Y - f(X))^2$$

$$R(f) = \mathbb{E} \left[(Y - f(X))^2 \right]$$

- Other examples for regression: Huber loss
- For classification: 0/1, logistic, hinge, etc.

The Empirical Risk

- With the training data $\mathcal{D}_n$, estimate $f(x)$ by minimizing the empirical risk

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i))$$

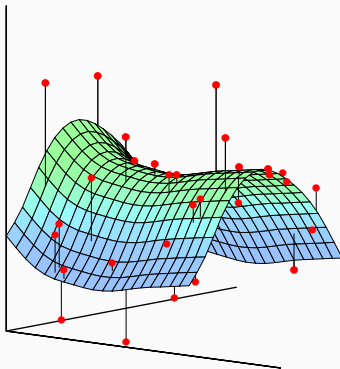
$$\hat{f}(x) = \arg \min_{f \in \mathcal{F}} R_n(f)$$

where $\mathcal{F}$ is some space of models

- Using the squared error loss for regression, we have

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2.$$

Regression



Estimating $f(X)$, figure from ESL

Linear Regression

- A **linear regression** model assumes a functional form of f

$$f(X) = X^T \beta$$

- Note: set $X_1 = 1$ as the intercept term.
- We express the regression problem in the **matrix form**

$$\mathbf{y}_{n \times 1} = \mathbf{X}_{n \times p} \boldsymbol{\beta}_{p \times 1} + \mathbf{e}_{n \times 1}$$

Linear Regression

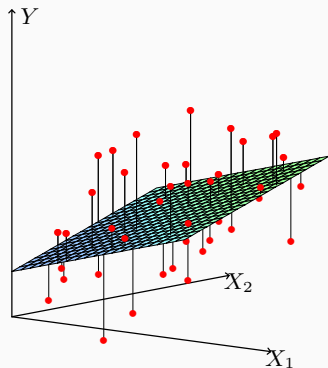
- Solve β by minimizing the residual sum of squares (**RSS**)

$$\begin{aligned}\text{RSS} &= \sum_{i=1}^n \left(y_i - x_{i1}\beta_1 - \cdots - x_{ip}\beta_p \right)^2 \\ &= \|\mathbf{y} - \mathbf{X}\beta\|^2 \\ &= (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)\end{aligned}$$

- The **ordinary least squares** estimator (**OLS**) is

$$\hat{\beta} = \arg \min_{\beta} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

Linear Regression



Solving linear regression, figure from ESL

Estimate β

- To estimate β , we set the derivative equal to 0

$$\begin{aligned}\frac{\partial \text{RSS}}{\partial \beta} &= -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) = 0 \\ \implies \mathbf{X}^\top \mathbf{y} &= \mathbf{X}^\top \mathbf{X} \beta\end{aligned}$$

which is the **normal equation**.

- We then have, if $\mathbf{X}^\top \mathbf{X}$ is invertible,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

- $\mathbf{X}$ full rank $\iff \mathbf{X}^\top \mathbf{X}$ invertible

Hat Matrix

- The fitted values (i.e., prediction at observed x_i 's) are

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \doteq \mathbf{H}_{n \times n} \mathbf{y}$$

- $\mathbf{H}$ ("hat matrix") is a **project** matrix
 - symmetric: $\mathbf{H}^\top = \mathbf{H}$
 - idempotent: $\mathbf{H}\mathbf{H} = \mathbf{H}$
- The **residual** $\mathbf{r}_{n \times 1} = \hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$
- $\mathbf{r}$ can be used to estimate the error variance

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum_{i=1}^n r_i^2 = \frac{\text{RSS}}{n-p}$$

- The essential machinery of linear regression is projection
- Decompose the outcome vector $\mathbf{y}$ into two orthogonal vectors

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{r}$$

- $\hat{\mathbf{y}}$ lives in the column space of $\mathbf{X}$, since $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$
- $\mathbf{r}$ is orthogonal to $\mathbf{X}$, i.e., $\mathbf{X}^T \mathbf{r} = \mathbf{0}$

Vector Space Interpretation

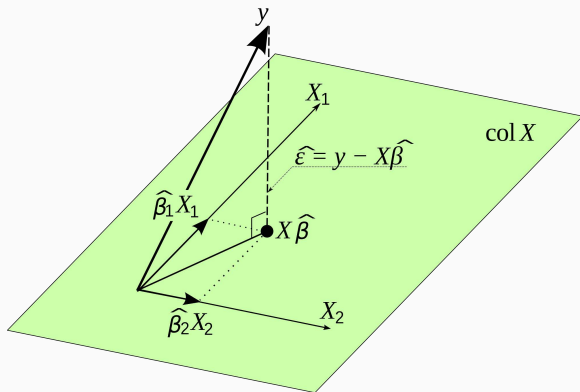


Figure from [Wiki](#)

Properties of $\hat{\beta}$

- We assume that the samples are generated from the model

$$Y = X^T \beta + \epsilon,$$

where the errors ϵ_i are i.i.d. with $E(\epsilon_i) = 0$ and $\text{Var}(\epsilon_i) = \sigma^2$

- Then $\hat{\beta}$ is unbiased: $E(\hat{\beta}) = \beta$
- Variance-covariance

$$\text{Var}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$$

- σ^2 can be estimated using $\hat{\sigma}^2$

Properties of $\hat{\beta}$

- Among all unbiased linear estimators, $\hat{\beta}$ has the smallest variance. (Gauss-Markov Theorem)
- An unbiased linear estimator is defined as

$$\hat{\beta} = \mathbf{A}\mathbf{y}, \text{ and } E(\hat{\beta}) = \beta$$

- Further assuming ϵ is normal, $\hat{\beta}$ is also UMVUE
- **Question:** What if we have a biased estimator? Can we trade a little bias for a large reduction in variance?

Training vs. Testing Errors

- In many applications nowadays, we have many explanatory variables, i.e., p is large or even $p \gg n$.
 - There are more than 20,000 human protein-coding genes
 - About 10 million single nucleotide polymorphisms (SNPs)
 - Number of subjects, n , is usually in hundreds or thousands
- In some applications, the key question is to identify a subset of X variables that are most relevant to Y
- Let's examine the training and testing errors from a linear model

Training vs. Testing error

- Training data $\mathcal{D}_n = \{x_i, y_i\}_{i=1}^n$
- Suppose $\{x_i, y_i^*\}_{i=1}^n$ is an independent (imaginary) testing dataset collected at the same location x_i 's (aka, in-sample prediction)
- Assume that the data are indeed from a linear model

$$\begin{aligned} \mathbf{y} &= \boldsymbol{\mu} + \mathbf{e} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \\ \mathbf{y}^* &= \boldsymbol{\mu} + \mathbf{e}^* = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}^* \end{aligned}$$

where both $\mathbf{y}$ and $\mathbf{y}^*$ are $n \times 1$ response vectors, $\mathbf{e}$ and $\mathbf{e}^*$ are i.i.d. error terms with mean 0 and variance σ^2 .

- The true model is indeed linear: $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$

Training vs. Testing error

$$\begin{aligned} E[\text{Test Err}] &= E\|\mathbf{y}^* - \mathbf{X}\hat{\beta}\|^2 \\ &= E\|(\mathbf{y}^* - \mathbf{X}\beta) + (\mathbf{X}\beta - \mathbf{X}\hat{\beta})\|^2 \\ &= E\|\mathbf{y}^* - \mu\|^2 + E\|\mathbf{X}(\hat{\beta} - \beta)\|^2 \\ &= E\|\mathbf{e}^*\|^2 + \text{Trace}(\mathbf{X}^\top \mathbf{X} \text{Cov}(\hat{\beta})) \\ &= n\sigma^2 + p\sigma^2 \end{aligned}$$

Training vs. Testing error

$$\begin{aligned} \mathbb{E}[\text{Train Err}] &= \mathbb{E}\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = \mathbb{E}\|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2 \\ &= \mathbb{E}\|(\mathbf{I} - \mathbf{H})\mathbf{e}\|^2 \\ &= \text{Trace}((\mathbf{I} - \mathbf{H})^\top (\mathbf{I} - \mathbf{H}) \text{Cov}(\mathbf{e})) \\ &= (n - p)\sigma^2 \end{aligned}$$

- Hence, the **testing error increases** with p and **training error decreases** with p . As p gets larger, this could be a big trouble...

Variable Selection

- It might be necessary to select a set of most relevant variables, especially when p is large.
- Variable selection may improve
 - Prediction accuracy
 - Interpretability
- This is a difficult task
 - No natural ordering of importance for the variables
 - The role of a variable needs be measured conditioning on others, high correlation causes trouble
 - It is essential to check all possible combinations, however, this may be computationally expensive

Model Selection Criteria and Algorithms

Model Selection Criteria

- Model selection is usually done in the following way
 - 1) Give each model a score
 - 2) Design an algorithm to find the model with the best (smallest) score
- The score of a model fitting takes the form

Goodness-of-fit + Complexity-Penalty

- 1) The first term decreases as the model gets more complicated (recall 1NN)
- 2) The second term increases with the number of predictor variables (recall degrees of freedom), which prefers “smaller” model

Model Selection Criteria

- Popular choices of scores:
 - Mallows' C_p (Mallows, 1973): $\text{RSS} + 2\hat{\sigma}_{\text{full}}^2 \cdot p$
 - AIC (Akaike, 1970): $-2 \text{ Log-likelihood} + 2 \cdot p$
 - BIC (Schwarz, 1978): $-2 \text{ Log-likelihood} + \log n \cdot p$
- When n is large, adding an additional predictor costs a lot more in BIC than AIC (or C_p). So AIC tends to pick a larger model than BIC.
- C_p performs similarly to AIC.

Justification of Mallows' C_p

- Recall our previous analysis of the training and testing errors with $\mathbf{y}$ and $\mathbf{y}^*$
- Now, let's assume that the model is not necessarily a linear model, i.e.,

$$\begin{aligned}\mathbf{y} &= \boldsymbol{\mu} + \mathbf{e} \\ \mathbf{y}^* &= \boldsymbol{\mu} + \mathbf{e}^*\end{aligned}$$

- We assume mean 0 and variance σ^2 for the two error vectors, but we don't have $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$. However, we still perform linear regression regardless. This will introduce bias of the estimations.

Justification of Mallows' C_p

$$\begin{aligned} E[\text{Test Err}] &= E\|\mathbf{y}^* - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 = \|\mathbf{y}^* - \mathbf{H}\mathbf{y}\|^2 \\ &= E\|(\mathbf{y}^* - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \mathbf{H}\boldsymbol{\mu}) + (\mathbf{H}\boldsymbol{\mu} - \mathbf{H}\mathbf{y})\|^2 \\ &= E\|\mathbf{y}^* - \boldsymbol{\mu}\|^2 + E\|\boldsymbol{\mu} - \mathbf{H}\boldsymbol{\mu}\|^2 + E\|\mathbf{H}\boldsymbol{\mu} - \mathbf{H}\mathbf{y}\|^2 \\ &= E\|\mathbf{e}^*\|^2 + E\|\boldsymbol{\mu} - \mathbf{H}\boldsymbol{\mu}\|^2 + E\|\mathbf{H}\mathbf{e}\|^2 \\ &= n\sigma^2 + \text{Bias}^2 + p\sigma^2 \end{aligned}$$

$$\begin{aligned} E[\text{Train Err}] &= E\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = E\|(\mathbf{I} - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I} - \mathbf{H})\mathbf{e}\|^2 \\ &= E\|(\mathbf{I} - \mathbf{H})\boldsymbol{\mu}\|^2 + E\|(\mathbf{I} - \mathbf{H})\mathbf{e}\|^2 \\ &= \text{Bias}^2 + (n - p)\sigma^2 \end{aligned}$$

Hence, **Test Err** is approximately **Train Err** + $2\sigma^2 p$, which justifies Mallows' C_p .

- To perform linear model selection, we need to decide on a selection criterion and use an computational algorithm to find the solution
 - Criteria: Mallows' C_p ; AIC; BIC
 - Algorithm: Best subset (Brute force); stepwise (forward/backward/...) selection
- Different algorithms may have different advantage

Best Subset Selection

- Best subset selection is a **level-wise search algorithm**, which returns the **global optimal** solution for a given model size.
- Only feasible for p not very large (< 50)
- Algorithm:
 - 1 For each $k = 1, \dots, p$, check $\binom{p}{k}$ possible combinations, and find the model with smallest RSS
 - The penalty term is the same for models with the same size
 - 2 To choose the best k , use model selection criteria

Best Subset Selection

- **Note:** if $\text{RSS}(X_1, X_2) < \text{RSS}(X_3, X_4, X_5, X_6)$ then we do not need to visit any size 2 or 3 sub-models of (X_3, X_4, X_5, X_6) , which can be **leaped** over.
- Implemented in **R** package **leaps**, using the leaps and bounds algorithm by Furnival and Wilson (1974)

Stepwise Regression

- **Greedy algorithms**: fast, but only return a local optimal solution (which might be good enough in practice).
 - **Backward**: start with the full model and sequentially delete predictors until the score does not improve.
 - **Forward**: start with the null model and sequentially add predictors until the score does not improve.
 - **Stepwise**: consider both deleting and adding one predictor at each stage.